



Quadratic equations: what we can already solve

WORKSHEET

Name _____

Class _____

Date _____

REMEMBER

From grade 8

Level 1 • threshold

1.Solve $2x^2 = 50$. Give both roots, separated by a semicolon.

Answer: _____

Level 1 • threshold

2.Solve $x^2 - 6x = 0$.

Answer: _____

Level 1 • threshold

3.Which of these numbers is a root of $x^2 - x - 12 = 0$?☐ A 3☐ B 12☐ C -4☐ D -3

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 1 • threshold

Write $4 - x^2 = 3x$ in the form $ax^2 + bx + c = 0$ with $a > 0$. Enter a , b and c , in this order, separated by semicolons.

Answer: _____

Example 2 • A product equal to zero: $(x - 4)(2x + 3) = 0$

ZERO-PRODUCT RULE

$$x - 4 = 0 \text{ or } 2x + 3 = 0.$$

SOLVE EACH

$$x = 4 \text{ or } 2x = -3, \text{ so } x = -\frac{3}{2}.$$

CHECK • YOUR TURN

YOUR TURN

Level 1 • threshold

Solve $(3x + 2)(x - 5) = 0$.

Answer:

Example 3 • A square equal to a number

$$(x + 2)^2 = 16$$

$$16 > 0: x + 2 = 4 \text{ or } x + 2 = -4, \text{ so } x = 2 \text{ or } x = -6.$$

$$(x + 4)^2 = 0 \text{ • YOUR TURN}$$

$$(x + 3)^2 = -4 \text{ • YOUR TURN}$$

YOUR TURN

Level 1 • threshold

Solve $(x - 3)^2 = 25$.

Answer:

CHECK

Check yourself

Level 1 • threshold

1.

Which of these are quadratic equations? Choose all that apply.

☐ $\frac{4}{x^2} - 1 = 0$

☐ $3^2x + x - 4 = 0$

☐ $x^2 - 7x = 0$

☐ $5 - 2x^2 = 0$

Level 2 • satisfactory

2.

Which of these is an **incomplete** quadratic equation?

☐ $x^2 + x = 1$

☐ $(x - 1)^2 = 0$

☐ $3x^2 = 5x$

☐ $2x^2 + 3x - 5 = 0$

3.

Write $7 - 3x^2 = 2x$ in the form $ax^2 + bx + c = 0$ with $a > 0$. Enter a , b and c , in this order.

Answer:

4.

Solve $(x + 6)(3x - 1) = 0$.

Answer:

5.

Solve $(x + 4)^2 = 36$.

Answer:

6.

Solve $(x - 5)^2 = -9$.

☐ A $x = -4$ ☐ B no real roots☐ C $x = 5$ ☐ D $x = 8$ or $x = 2$

7.

Ona solves $x(x - 3) = 4$ like this: " $x = 4$ or $x - 3 = 4$, so $x = 4$ or $x = 7$." What is wrong?

☐ A The zero-product rule needs 0 on one side. Here it is 4, so the rule does not apply, and $x = 7$ is not a root.☐ B She should divide both sides by x .☐ C Nothing: 4 and 7 are both roots.☐ D She should have written $x = 0$ or $x - 3 = 0$.

EXAM TASK

A hidden square

Level 3 · basic

PUPP style · 3 points

The equation $x^2 - 2x + 1 = k$ is given, where k is a number.

- a) Write the left side as the square of a binomial.
- b) Solve the equation when $k = 49$.
- c) For which values of k does the equation have no real roots? Explain.

